3.1. Notes

Site: Eduvos LMS
Course: ITNMA4 (First Block)
Book: 3.1. Notes
Printed by: Sibabalwe Mfisa
Date: Wednesday, 15 July 2026, 6:12 PM

Table of contents

1. Learning outcomes
2. 5G Architectural Components
- 2.1. Access Network
- 2.2. Packet Core Network
- 2.3. Transport Network
- 2.4. Practice Activity

1. Learning outcomes

By the end of this lesson, you should be able to:

- Examine 5G concepts and architecture.

Prescribed Reading

Shyam Varan Nath, Ananya Simlai & Oğuzhan Kara. Mastering 5G Network Design, Implementation, and Operations: A comprehensive guide to understanding, designing, deploying, and managing 5G networks. 1st Edition. 2023. Packet Publishing.

Chapter 2

Open book in new window

Not signed in? Click here and then refresh this page.

Need help? Contact Support

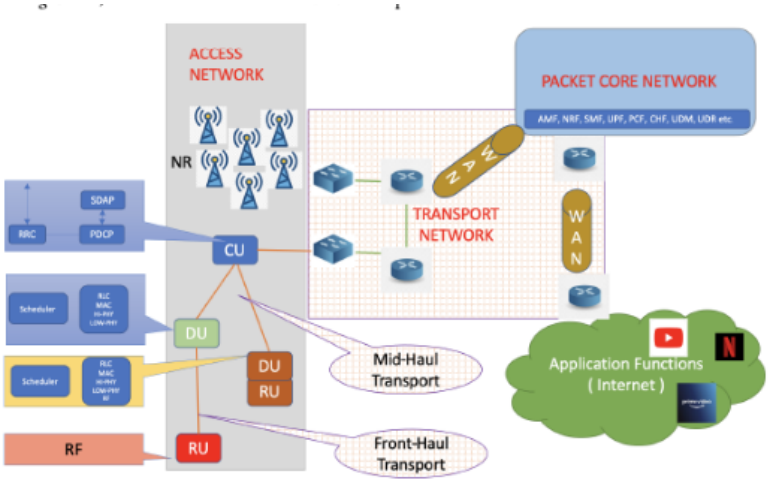
Please note: You will only be able to access the book on Kortext if you have purchased it with your vossie.net account via the Eduvos eBookstore.

2. 5G Architectural Components

A 5G network has three architectural components which are.

1. Access network
2. Packet core network
3. Transport network.

The components are shown in the diagram below.



2.1. Access Network

Access Network

Access network comprises the 5G radio components i.e. the New Radio.

- It is responsible for the transmission of radio waves to the User Equipment like handsets, sensors and IoT equipment.
- It is further divided into the following components:

    1. Centralized Unit (CU),
    2. Distributed Unit (DU),
    3. Radio Unit (RU).

The 5G access network is a critical enabler for the broad range of applications and services that 5G technology promises to deliver. It is continuously evolving to meet the increasing demands of mobile broadband, IoT, and emerging use cases in the era of digital connectivity.

2.2. Packet Core Network

Packet core Network

- The 5G Packet Core Network is a key architectural component of (5G) wireless networks, which provides the core infrastructure for managing and routing data traffic between end-user devices and external networks.
- It is an evolution of the packet core networks found in earlier generations like 4G LTE.
- It comprises control and User Plane Functions (UPFs) that are responsible for the management of the subscriber session, charging, data processing, and Quality of Service (QoS).
- Other functionalities include:

    1. Session Management Function (SMF).
    2. Access and Mobility Management Function (AMF)
    3. Unified Data Management (UDM)
    4. Authentication Server Function (AUSF)
    5. Network Exposure Function (NEF)
    6. Policy Control Function (PCF
    7. Network Slice Selection Function (NSSF)

2.3. Transport Network

Provides connectivity between packet core network and user access network. It provides high-speed, low-latency connectivity between various components of the 5G architecture. Below are the key aspects and functionalities of the transport network ;

1. High Capacity and Low Latency:

- The transport network in 5G is designed to handle significantly higher data rates and lower latency compared to previous generations. This is crucial for supporting advanced applications like augmented reality (AR), virtual reality (VR), and real-time communications.

2. Network Slicing:

- Network slicing is a fundamental concept in 5G, allowing the transport network to be partitioned into multiple virtual networks or slices. Each slice can be optimized for specific use cases, providing customized connectivity, bandwidth, and latency characteristics.

3. Fronthaul and Backhaul Networks:

- The transport network includes both fronthaul and backhaul segments. Fronthaul connects the radio units (RUs) to the central unit (CU) in the RAN, while backhaul connects the core network to the RAN. These segments collectively ensure seamless communication between the radio access points and the core network.

4. Flexibility in Access Technologies:

- The 5G transport network accommodates a variety of access technologies, including fiber optics, microwave, and millimeter-wave wireless links. This flexibility allows operators to deploy transport solutions based on the specific characteristics of the deployment area.

5. Edge Computing Integration:
   - To support low-latency applications and services, the transport network in 5G integrates edge computing capabilities. Edge nodes are strategically placed at the edge of the network to process data closer to the end-user, reducing latency and enhancing the overall user experience.

6. Carrier Aggregation and Massive MIMO Support:
   - The transport network supports technologies like carrier aggregation and Massive MIMO (Multiple-Input, Multiple-Output) to enhance spectral efficiency and boost data rates. These technologies are vital for optimizing the utilization of available radio frequencies.

7. Time-Sensitive Networking (TSN):
   - 5G transport networks employ Time-Sensitive Networking (TSN) standards to ensure precise and deterministic timing. This is crucial for applications that require synchronized communication and coordination, such as industrial automation.

8. Security Considerations:
   - Security is a critical aspect of the 5G transport network. It employs encryption, secure key exchange mechanisms, and authentication protocols to protect data in transit from potential security threats.

9. Orchestration and Automation:
   - The transport network is orchestrated and automated to dynamically adapt to changing network conditions, traffic patterns, and service requirements. This ensures efficient resource utilization and enhances the overall network performance.

2.4. Practice Activity

Write a short essay (200–300 words) elaborating the relationship between the Access Network, Packet Core Network, and Transport Network in enabling seamless connectivity for end users. Include specific examples from the rural and urban deployment scenarios provided.